

KNOWLEDGE SCHEMA

Purpose: Define the structure and types for the knowledge graph that ensures consistency across the book
Last Updated: January 11, 2026
Version: 2.0 (Added definition_type and related fields)

SCHEMA OVERVIEW

This schema defines how we structure extracted entities and relationships to build a reliable knowledge graph. Consistency here enables:

- Accurate content generation
- Cross-chapter consistency
- Reliable validation
- No contradictory definitions

CORE ENTITY SCHEMA

Base Entity Structure

All entities include these base fields:

```
{
  "name": "string",           // Entity name (e.g., "The Ring", "System of
                              // Systems")
  "type": "string",           // Entity type (see Entity Types below)
  "definition": "string",     // Precise definition
  "definition_type": "string", // NEW: "technical_standard" | "book_coined" |
  "book_original"
  "first_mention": "string",   // Where first introduced (e.g., "Chapter 1,
  "context": "string",         // Brief context of introduction
  "related_entities": ["array"] // Names of related entities
}
```

THREE-TIER DEFINITION PROTOCOL

Type 1: Technical Standard

For established academic/technical terms

Required Fields:

```
{
  "name": "System of Systems",
  "type": "framework",
  "definition": "A collection of task-oriented or dedicated systems that pool their resources and capabilities together to create a new, more complex system which offers more functionality and performance than simply the sum of the constituent systems",
  "definition_type": "technical_standard",
  "source": "Systems engineering",
  "book_usage": "Applied to political ideologies as constituent systems operating at different scales within a larger system of human relationships",
  "first_mention": "Prologue, paragraph 3",
  "context": "Introduced after demonstrating slug lines cannot be purely categorized",
  "related_entities": ["Slug Line", "Four Habitats", "Scale"]
}
```

Type-Specific Fields:

- **source** (required): Field/discipline of standard definition
- **book_usage** (required): How this book applies the technical term

Type 2: Book-Coined

For terms Joshua defines specifically for the book

Required Fields:

```
{
  "name": "Scale Errors",
  "type": "concept",
  "definition": "The mistake of applying a system designed for one scale of human relationship to a different scale where it doesn't fit",
  "definition_type": "book_coined",
  "builds_on": "Common understanding of 'scale' and 'error' but compound meaning is book-specific",
  "first_mention": "Chapter 1, title and opening",
  "context": "Central concept explaining why ideological fights happen",
  "examples": ["Forcing family to operate like market", "Applying market logic to public safety"],
  "related_entities": ["Four Habitats", "The Ring", "System of Systems"]
}
```

Type-Specific Fields:

- **builds_on** (optional): Common usage the term builds on

Type 3: Book-Original

For Joshua's completely original frameworks

Required Fields:

```
{
  "name": "The Ring",
  "type": "framework",
  "definition": "A circular model showing four overlapping systems arranged around a center, illustrating continuous movement through all four rather than choosing one ideology",
  "definition_type": "book_original",
  "components": ["Family (communism)", "Safety Commons (socialism)", "Markets (capitalism)", "Voluntary Networks (anarchy)"],
  "purpose": "Replaces the broken left/right spectrum with a multi-dimensional navigation tool",
  "visual_representation": "Circle/ring diagram with four quadrants",
  "first_mention": "Chapter 1, after establishing scale errors",
  "replaces": "Left/Right political spectrum",
  "related_entities": ["Four Habitats", "Scale Errors", "Left/Right Spectrum"],
  "used_throughout": true
}
```

Type-Specific Fields:

- **components** (optional): Parts/elements of the framework
- **purpose** (required): What problem this framework solves
- **visual_representation** (optional): Description of diagram/visual
- **replaces** (optional): What framework this supersedes

ENTITY TYPES

1. CORE_CONCEPTS

Types:

- **framework**: Structured models or systems (The Ring, Four Habitats)
- **concept**: Abstract ideas or principles (Scale Errors, Emergence)
- **principle**: Rules or patterns (Obligation to Excess)
- **mechanism**: Process descriptions (Surplus → Excess → Discovery)

Schema:

```
{
  "name": "string",
  "type": "framework | concept | principle | mechanism",
  "definition": "string",
  "definition_type": "technical_standard | book_coined | book_original",

  // Type-specific fields based on definition_type (see above)
```

```

    "first_mention": "string",
    "context": "string",
    "related_entities": ["array"],
    "evolution_notes": "string (optional)"
  }

```

2. EXAMPLES

Types:

- **story_example**: Narrative examples (Slug Line, Maya's Hospital)
- **historical_reference**: Historical events/figures (Nietzsche's horse)
- **metaphor**: Comparative illustrations
- **real_world_application**: Practical applications

Schema:

```

{
  "name": "string",
  "type": "story_example | historical_reference | metaphor |
real_world_application",
  "description": "string",
  "illustrates": ["array"],      // Concepts this example demonstrates
  "location": "string",         // Where in manuscript
  "usage_pattern": "recurring | single_use | primary_example",
  "related_entities": ["array"],
  "narrative_arc": "string (optional)" // If story has progression
}

```

Example:

```

{
  "name": "Slug Line",
  "type": "story_example",
  "description": "Washington DC's self-organizing carpool system where strangers
share rides to cross HOV lanes",
  "illustrates": ["Four Habitats", "Emergent Order", "Anarchy", "Voluntary
Networks"],
  "location": "Prologue, opening story",
  "usage_pattern": "primary_example",
  "related_entities": ["Four Habitats", "Anarchy", "Emergent Order", "System of
Systems"],
  "narrative_arc": "Used throughout book to demonstrate all four habitats"
}

```

3. DEFINITIONS

For terms needing disambiguation or precise usage

Schema:

```
{
  "term": "string",
  "definition": "string",
  "definition_type": "technical_standard | book_coined | book_original",
  "usage_context": "string",
  "not_to_be_confused_with": "string (optional)",

  // Type-specific fields based on definition_type
  "source": "string (if technical_standard)",
  "builds_on": "string (if book_coined)",
  "purpose": "string (if book_original)"
}
```

Example:

```
{
  "term": "Habitat",
  "definition": "A scale of human relationship where a particular system operates naturally",
  "definition_type": "book_coined",
  "builds_on": "Biological sense of 'habitat' as natural environment",
  "usage_context": "Always used with 'Four Habitats' framework",
  "not_to_be_confused_with": "Literal biological habitat"
}
```

RELATIONSHIPS SCHEMA

Relationship Types:

- **contains**: A contains B (The Ring contains Four Habitats)
- **depends_on**: A requires understanding B (Scale Errors depends on Four Habitats)
- **contrasts_with**: A vs. B (Left/Right Spectrum vs. The Ring)
- **progresses_to**: A leads to B (Surplus → Excess)
- **illustrates**: Example demonstrates concept (Slug Line illustrates Four Habitats)
- **builds_on**: A extends B (book concept builds on technical term)

Schema:

```
{
  "entity_1": "string",           // Name of first entity
```

```

"relationship_type": "contains | depends_on | contrasts_with | progresses_to |
illustrates | builds_on",
"entity_2": "string",           // Name of second entity
"description": "string",       // Brief explanation
"directional": true/false,     // Is this one-way or bidirectional?
"strength": "strong | moderate | weak (optional)"
}

```

Examples:

```

{
  "entity_1": "The Ring",
  "relationship_type": "contains",
  "entity_2": "Four Habitats",
  "description": "The Ring framework consists of four habitat quadrants",
  "directional": true,
  "strength": "strong"
}

```

```

{
  "entity_1": "Scale Errors",
  "relationship_type": "depends_on",
  "entity_2": "Four Habitats",
  "description": "Understanding scale errors requires knowing which habitats
exist",
  "directional": true,
  "strength": "strong"
}

```

```

{
  "entity_1": "Slug Line",
  "relationship_type": "illustrates",
  "entity_2": "Four Habitats",
  "description": "Slug line demonstrates all four habitats in action",
  "directional": true,
  "strength": "strong"
}

```

EXTRACTION OUTPUT SCHEMA

Complete JSON structure for each extracted file:

```

{
  "source_file": "string",           // Path to manuscript file

```

```

"extraction_date": "YYYY-MM-DD",
"extractor": "opus-4.5 | sonnet-4.5",
"batch": "number (optional)",

"entities": {
  "core_concepts": [
    {
      // Core concept schema (see above)
    }
  ],
  "examples": [
    {
      // Example schema (see above)
    }
  ],
  "definitions": [
    {
      // Definition schema (see above)
    }
  ]
},

"relationships": [
  {
    // Relationship schema (see above)
  }
],

"extraction_notes": {
  "ambiguities": ["array"], // Unclear definitions or usage
  "evolution_tracking": ["array"], // Terms that may evolve later
  "cross_references": ["array"], // References to other chapters
  "quality_concerns": ["array"], // Any extraction issues
  "technical_terms_verified": true/false // Did we check standard definitions?
}
}

```

MASTER DATABASE SCHEMA

Consolidated entity database structure:

File: <reference/knowledge-graph/entity-database.json>

```

{
  "metadata": {
    "last_updated": "YYYY-MM-DD",
    "total_entities": "number",
    "total_relationships": "number",
    "sources": ["array of manuscript files"],
    "schema_version": "2.0"
  }
}

```

```

    },

    "entities": {
      "core_concepts": {
        "[entity_name]": {
          // Full entity object
          "appearances": [
            {
              "file": "string",
              "context": "string",
              "evolution": "string (if definition evolves)"
            }
          ]
        }
      },
    },
    "examples": {
      "[example_name]": {
        // Full example object
        "appearances": ["array"]
      }
    },
    "definitions": {
      "[term]": {
        // Full definition object
      }
    },
  },

  "relationships": [
    // Array of all relationships
  ],

  "indexes": {
    "by_type": {
      "framework": ["array of names"],
      "concept": ["array of names"],
      // etc.
    },
    "by_chapter": {
      "prologue": ["array of entities"],
      "chapter-01": ["array of entities"],
      // etc.
    },
    "by_definition_type": {
      "technical_standard": ["array"],
      "book_coined": ["array"],
      "book_original": ["array"]
    }
  }
}

```


VALIDATION RULES

Required Field Validation

All entities must have:

- **name** (string, non-empty)
- **type** (valid entity type)
- **definition** (string, non-empty, NOT circular)
- **definition_type** (valid type: technical_standard | book_coined | book_original)

Type-specific requirements:

If **definition_type**: "technical_standard":

- Must have **source** field
- Must have **book_usage** field
- Definition should be from recognized source, not improvised

If **definition_type**: "book_coined":

- May have **builds_on** field
- Definition should be Joshua's exact wording

If **definition_type**: "book_original":

- Must have **purpose** field
- May have **components** array

Circular Definition Check

Invalid (REJECT):

```
{
  "name": "System of Systems",
  "definition": "A system of systems is..."
}
```

Valid:

```
{
  "name": "System of Systems",
  "definition": "A collection of task-oriented or dedicated systems that pool
their resources..."
}
```

Rule: Definition text must NOT contain the entity name itself.

Consistency Checks

Across files:

- Same entity name = same definition
 - If definition evolves, track in [evolution_notes](#)
 - Relationships must reference existing entities
 - No orphaned entities (must relate to something)
-

USAGE EXAMPLES

Example 1: Technical Term

```
{
  "name": "Nash Equilibrium",
  "type": "concept",
  "definition": "A solution concept in game theory where each player's strategy is optimal given the strategies of all other players, meaning no player can improve their outcome by unilaterally changing strategy",
  "definition_type": "technical_standard",
  "source": "Game theory (economics/mathematics)",
  "book_usage": "Used to explain why purely self-interested behavior in slug lines doesn't lead to chaos but to stable cooperation",
  "first_mention": "Chapter 6, when explaining emergent order",
  "context": "Demonstrates that anarchy doesn't mean disorder",
  "related_entities": ["Slug Line", "Emergent Order", "Voluntary Networks", "Game Theory"]
}
```

Example 2: Book-Coined Term

```
{
  "name": "Witching Hour",
  "type": "concept",
  "definition": "The time when HOV lane restrictions lift and the incentive for drivers to pick up extra passengers disappears, causing the slug line system to dissolve until the next day",
  "definition_type": "book_coined",
  "builds_on": "Folk term 'witching hour' meaning a time of magical transformation",
  "first_mention": "Prologue, after describing slug line operations",
  "context": "Sets up the puzzle of why the driver picked up the author after the advantage was gone",
  "examples": ["Slug Line narrative"],
  "related_entities": ["Slug Line", "Homo Economicus", "Obligation to Excess"]
}
```

Example 3: Book-Original Framework

```
{
  "name": "Four Habitats",
  "type": "framework",
  "definition": "Four distinct scales of human relationship (Family, Safety Commons, Markets, Voluntary Networks) where different organizing systems (communism, socialism, capitalism, anarchy) operate naturally and effectively",
  "definition_type": "book_original",
  "components": [
    "Family - intimate scale (communism)",
    "Safety Commons - community scale (socialism)",
    "Markets - population scale (capitalism)",
    "Voluntary Networks - flexible scale (anarchy)"
  ],
  "purpose": "Explains that political ideologies are not competing philosophies but tools optimized for different scales of human organization",
  "visual_representation": "Four quadrants within The Ring",
  "first_mention": "Chapter 1, after establishing scale errors",
  "related_entities": ["The Ring", "Scale Errors", "Communism", "Socialism", "Capitalism", "Anarchy"],
  "used_throughout": true
}
```

SCHEMA EVOLUTION

Version History:

v1.0 (Initial):

- Basic entity structure
- Simple relationships

v2.0 (Current - January 11, 2026):

- Added `definition_type` field
- Added type-specific fields:
 - `source` and `book_usage` for `technical_standard`
 - `builds_on` for `book_coined`
 - `components` and `purpose` for `book_original`
- Enhanced validation rules
- Added circular definition prevention

Future Considerations:

- Versioning for evolving definitions
- Cross-book references (if series)
- Reader feedback integration

- Visual relationship mapping
-

INTEGRATION WITH OTHER SYSTEMS

This schema feeds:

1. Content Generation (Sonnet 4.5):

- Provides exact definitions to use
- Ensures terminology consistency
- Supplies examples for integration

2. Validation (Opus 4.5):

- Reference for checking consistency
- Definition verification
- Relationship validation

3. Claude.ai Project:

- Progress tracking
- Continuity monitoring
- Cross-chapter consistency

4. Knowledge Graph Visualization:

- Node structure
 - Edge relationships
 - Metadata for display
-

This schema is the authoritative structure. All entity extraction must conform to these specifications.